Course Logistics

CMSC477 – Robotics: Perception & Planning
Spring 2026

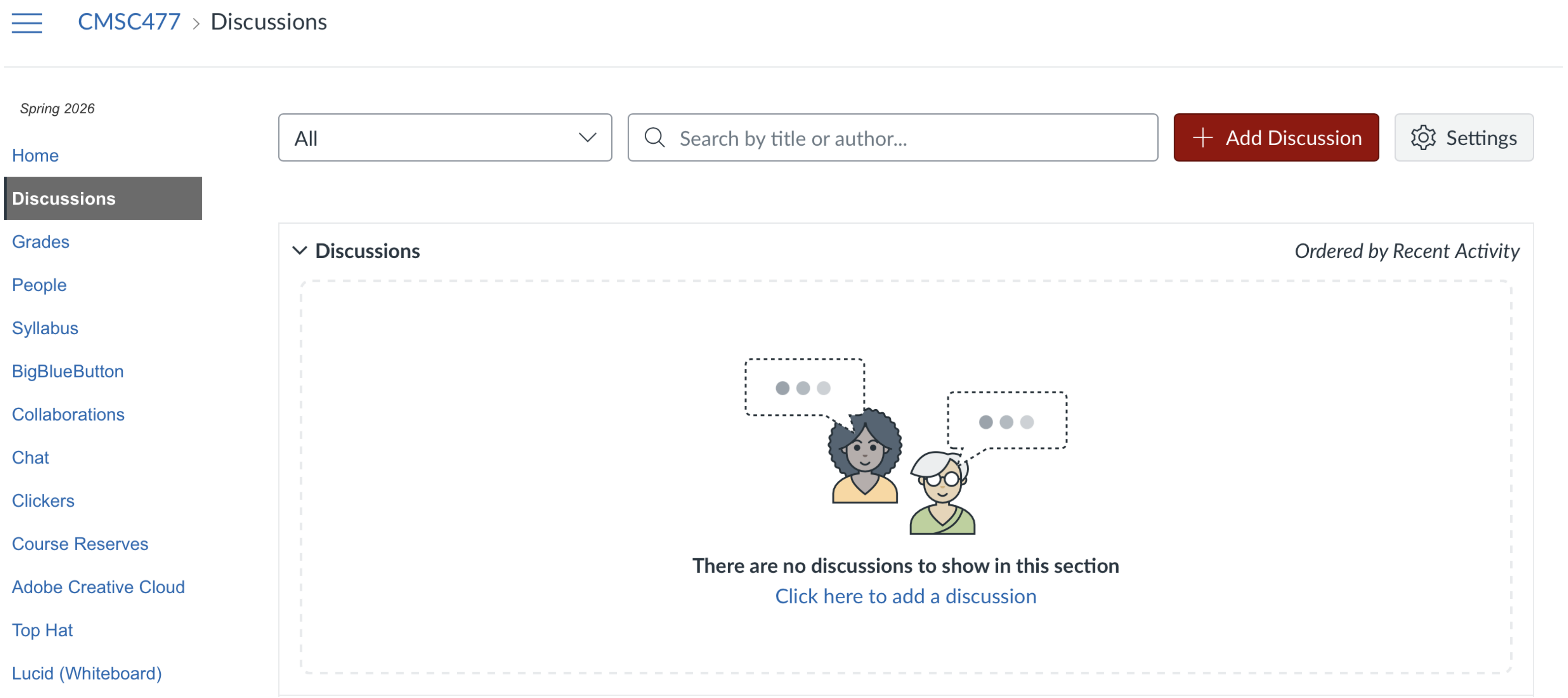
TAs: Anukriti Singh, Khuzema Habib

Lab Manager: Ivan Pensiky

Agenda

- Course Website/ELMS
- Lab Logistics
- Robots
- Getting Started: Project Zero
- Homework 1

Discussions on ELMS



Labs

- Lab, 3rd Floor of IDEA Factory
  - Managed by Ivan Pensiky
- Official Lab Hours (attendance mandatory)
  - Section 0101: Monday 2:00 - 4:50 pm
  - Section 0202: Tuesday 2:30 - 5:20 pm

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- ***Who is YOUR TA?***

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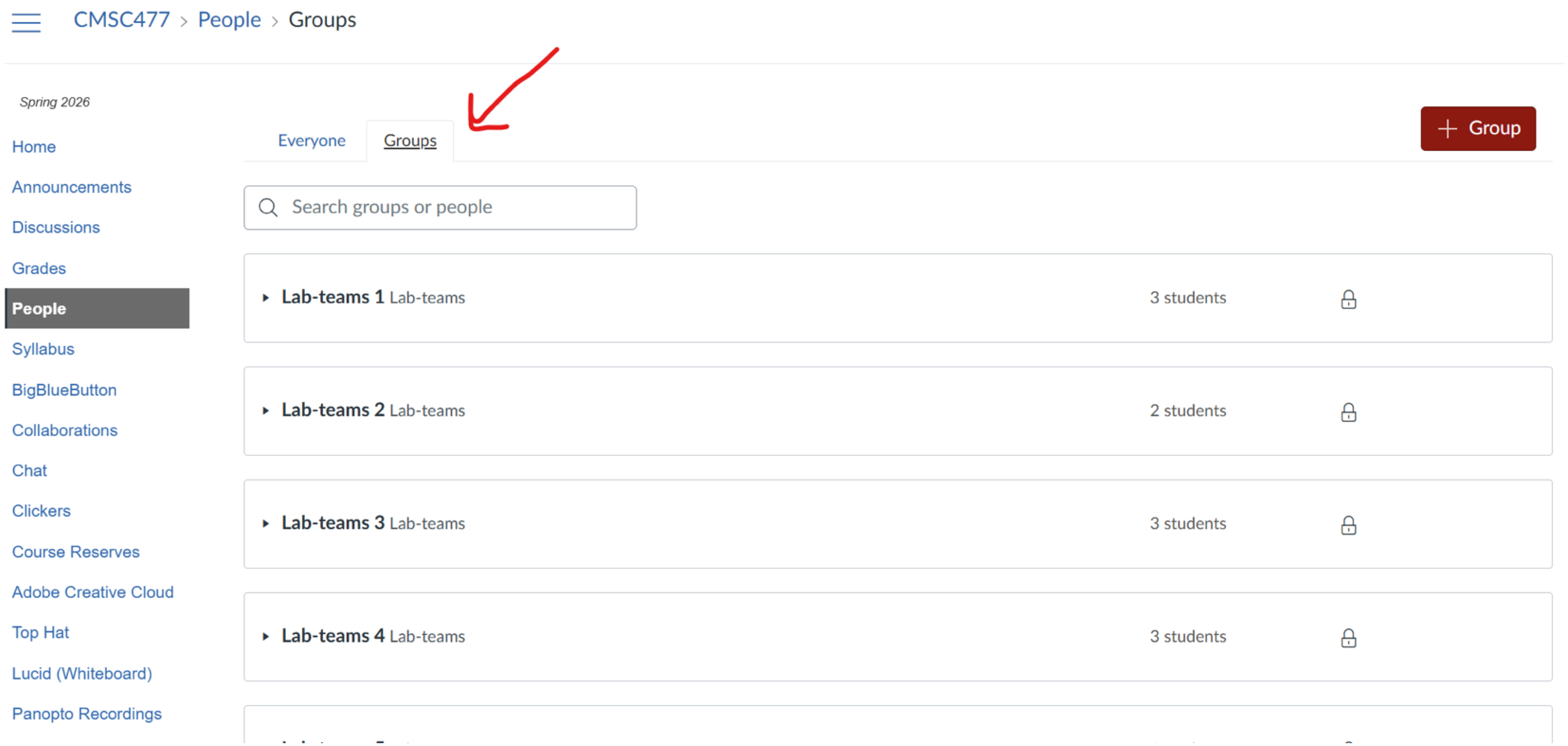
- Official Lab Hours (attendance mandatory)
  - Section 0101: Monday 2:00 - 4:50 pm -– **Khuzema Habib**
  - Section 0202: Tuesday 2:30 - 5:20 pm -– **Anukriti Singh**

- *Direct any questions about the assignments, labs, and robots to your TA*!

Team Assignments

- List is be posted on ELMS



Team Assignments

- List is be posted on ELMS
- You are free to switch teams
  - As long as number of teams does not change!
  - Let Anukriti know about any changes by Friday, Feb 5—*otherwise, you will remain with the same individuals you were assigned*!

- One (1) robot per team
  - Your team’s robot id numbers are on the sheet

End of Semester Competition Date

- May 6th is problematic.

- Other options for competition dates:
  - May 4
  - Final Exam day (May 14)

Let’s Meet Your TAs

Khuzema Habib --- Mondays (Section 0101)

Anukriti Singh --- Wednesdays (Section 0102)

Robots

- Each team has one (1) DJI RoboMaster
    - Each robot has 2 batteries
- Camera
- 2 axis Gripper
- Mecanum wheels
    - Can rotate and translate independently
- Communicates via WiFi

Robots

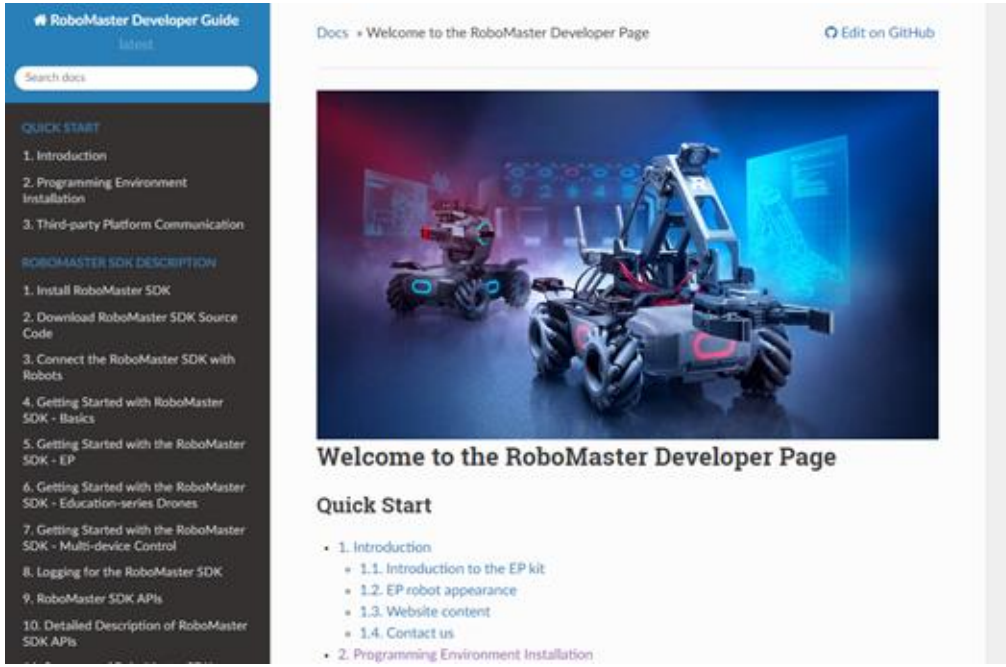
- Python code runs on your laptop
    - Receives camera images
    - Sends motor commands
- Communication occurs over WiFi

Logistics

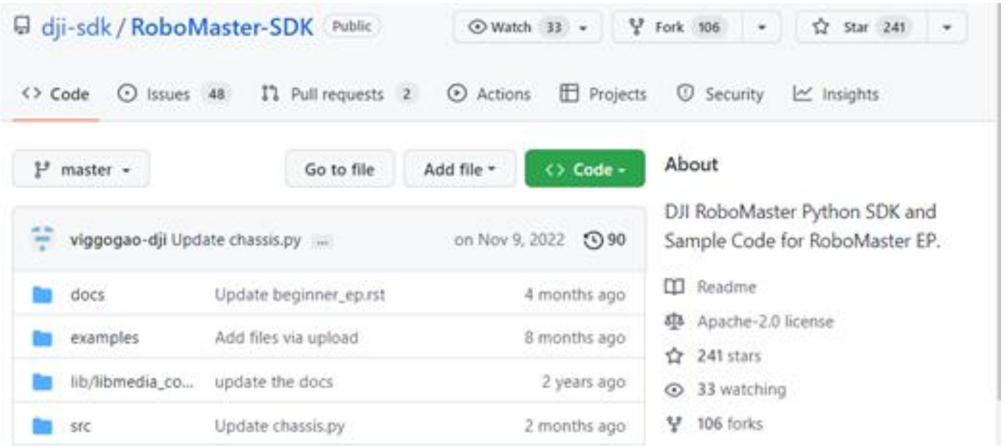
- You ***must*** work with your robots in the lab ***only.***
- Be very **careful** with them!
  - They are expensive and hard to replace.
- **Do not** use them on a table.
- **Do not** crash your robot into things!

RoboMaster SDK

- Python Library provided by DJI
- Documentation
    - https://robomaster-dev.readthedocs.io/en/latest/

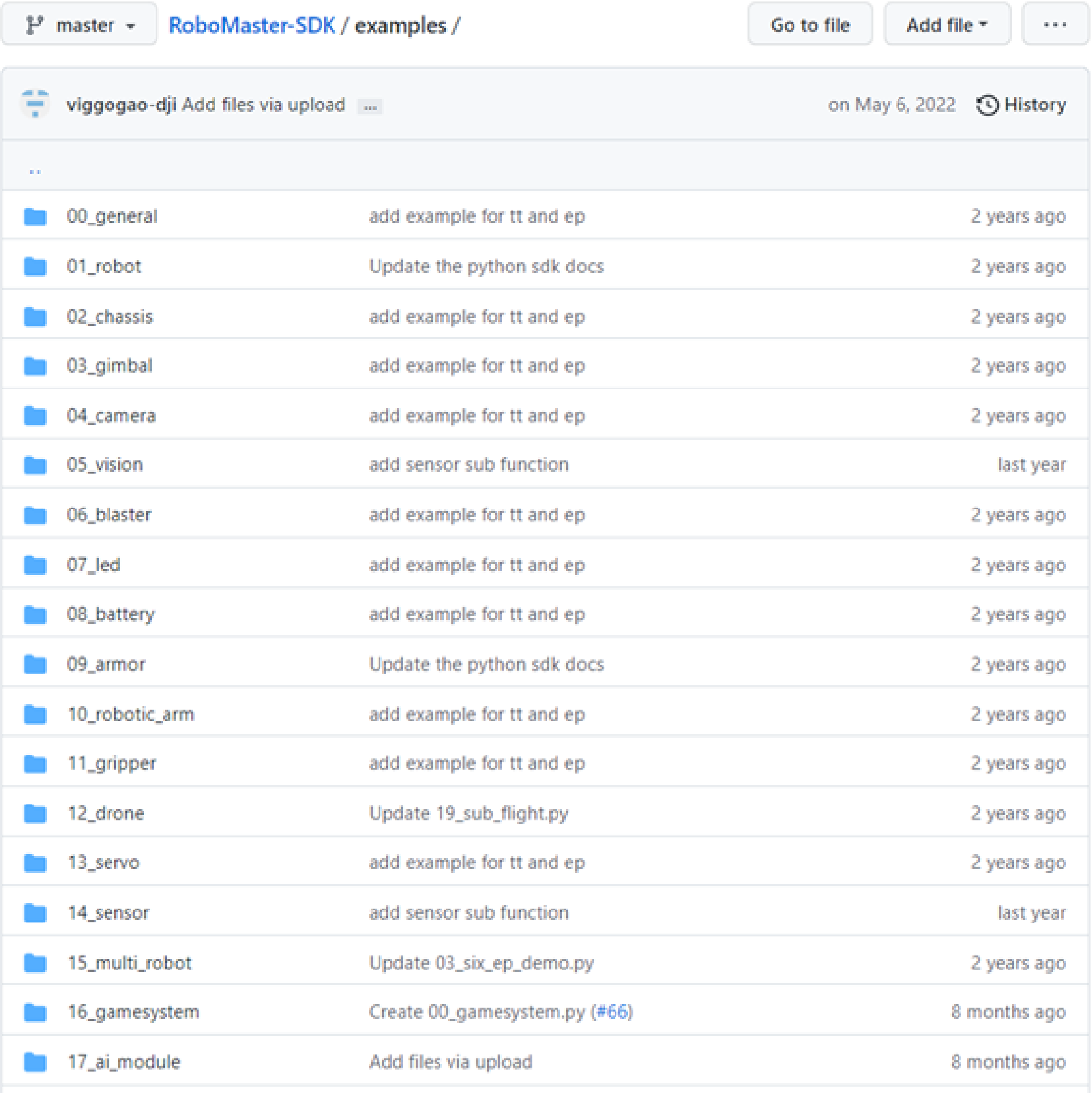


- Github Repository
    - https://github.com/dji-sdk/RoboMaster-SDK



RoboMaster SDK

- Numerous Python examples



master | RoboMaster-SDK / examples / | Go to file | Add file | ...

viggogao-dji Add files via upload ... on May 6, 2022 History

..		
00_general	add example for tt and ep	2 years ago
01_robot	Update the python sdk docs	2 years ago
02_chassis	add example for tt and ep	2 years ago
03_gimbal	add example for tt and ep	2 years ago
04_camera	add example for tt and ep	2 years ago
05_vision	add sensor sub function	last year
06_blaster	add example for tt and ep	2 years ago
07_led	add example for tt and ep	2 years ago
08_battery	add example for tt and ep	2 years ago
09_armor	Update the python sdk docs	2 years ago
10_robotic_arm	add example for tt and ep	2 years ago
11_gripper	add example for tt and ep	2 years ago
12_drone	Update 19_sub_flight.py	2 years ago
13_servo	add example for tt and ep	2 years ago
14_sensor	add sensor sub function	last year
15_multi_robot	Update 03_six_ep_demo.py	2 years ago
16_gamesystem	Create 00_gamesystem.py (#66)	8 months ago
17_ai_module	Add files via upload	8 months ago

RoboMaster SDK

- Simple API

```python
import time
from robomaster import robot


if __name__ == '__main__':
    ep_robot = robot.Robot()
    ep_robot.initialize(conn_type="sta")

    ep_chassis = ep_robot.chassis

    x_val = 0.5
    y_val = 0.3
    z_val = 30

    # 前进 3秒
    ep_chassis.drive_speed(x=x_val, y=0, z=0, timeout=5)
    time.sleep(3)

    # 后退 3秒
    ep_chassis.drive_speed(x=-x_val, y=0, z=0, timeout=5)
    time.sleep(3)

    # 左移 3秒
    ep_chassis.drive_speed(x=0, y=-y_val, z=0, timeout=5)
    time.sleep(3)

    # 右移 3秒
    ep_chassis.drive_speed(x=0, y=y_val, z=0, timeout=5)
    time.sleep(3)

    # 左转 3秒
    ep_chassis.drive_speed(x=0, y=0, z=-z_val, timeout=5)
    time.sleep(3)

    # 右转 3秒
    ep_chassis.drive_speed(x=0, y=0, z=z_val, timeout=5)
    time.sleep(3)

    # 停止麦轮运动
    ep_chassis.drive_speed(x=0, y=0, z=0, timeout=5)
```

Project 0

- Description posted today on website
- Track an AprilTag that is moving
  - Remain a fixed distance away
- This challenge is just to get you started
- Video due: **February 20th**
  - *Two* full lab sessions till due date
  - Later projects will have written report

Homework 1

- One and only homework in this class
- Will be released by February 10th

CMSC477: Robotics Perception and Planning
Homework 1: Graph-Based Planning

Dr. Troi Williams, Anukriti Singh, Khuzema Habib

February 10, 2026

1 Introduction

In this assignment, you will learn to find the shortest valid path for a robot to move from a source to a destination, given a series of obstacles. You will be solving this problem using Graph-Based Search Algorithms:

- Breadth-First Search (BFS)
- Depth-First Search (DFS)
- Djikstra

Questions?